Brunda Marpadaga

Embedded Firmware Engineer | C/C++ | FreeRTOS | Embedded Linux | FPGA SoC Bring-up
[redacted] | brundamarpadaga@gmail.com | LinkedIn | GitHub | Portfolio

Education

Portland State University Portland, OR
M.S. Electrical and Computer Engineering – ***GPA 3.85/4*** *Graduated Dec 2025*

- **Coursework:** Embedded Systems Programming, Microprocessor System Design, Computer Architecture

Technical Skills

Languages: C, C++, SystemVerilog, MATLAB
Firmware & OS: FreeRTOS, bare-metal, embedded Linux (PetaLinux/Yocto, device tree)
Interfaces: I2C, SPI, UART, GPIO, AXI, Ethernet, TCP/IP, MQTT
Hardware: STM32, ESP8266/Arduino, Xilinx Zynq-7000 SoC (Zybo Z7-20), Artix-7 (Nexys A7-100T)
Debug & Tools: JTAG, J-Link/Segger Ozone, GDB, STM32CubeIDE, Vitis, CMake, Git, oscilloscope, logic analyzer

Projects

FPGA Accelerator + Dual-Core Firmware on Zynq SoC | *C, FreeRTOS, ARM Cortex-A9, DMA* code

- Built the firmware and tests that drive an FPGA matrix-multiply accelerator from an ARM processor, running up to 9.4× faster than software baseline.
- Implemented asymmetric multiprocessing (AMP) running FreeRTOS on core 0 and bare-metal C on core 1, with separate memory regions, shared-memory messaging, and a lock for the shared UART port.
- Profiled DMA data paths across coherent (ACP) and non-coherent (HP) AXI ports, proving hardware coherency removes software cache invalidation overhead for a 1.9–2.6× throughput gain.

Embedded Linux Sensor Node with Cloud Telemetry | *C, PetaLinux/Yocto, Device Tree, AXI, I2C* code

- Brought up PetaLinux on a Zybo Z7-20 board, integrating PL peripherals (Xilinx IIC and GPIO) and debugging device tree bindings to probe standard Linux kernel drivers.
- Wrote user-space C drivers for a temperature/humidity sensor and an OLED display, and added an MQTT/TLS client to the build to send readings to the cloud.
- Fixed an I2C driver that wouldn't load (interrupt missing from the device tree), a math overflow in sensor readings, and TLS failures caused by the board's missing clock.

FreeRTOS Light Controller with I2C Sensor Driver | *C, FreeRTOS, I2C, PWM, MicroBlaze*

- Wrote an I2C driver for a light sensor from its datasheet, and used FreeRTOS tasks with PID control and PWM to hold an LED at a set brightness.
- Built the soft-core processor system on an Artix-7 FPGA and debugged it over JTAG.

IoT Infant Monitor | *C++, Arduino ESP8266, Wi-Fi*

- Read 3 sensors on an ESP8266 using hardware interrupts, drove a servo, and sent data over Wi-Fi with phone alerts.

Experience

Android Developer (Open Source Contributor) Mar 2026 – Present
Greenstand *Remote, U.S.*

- Built mobile features in Kotlin/Compose for the open-source Treetracker app, collaborating via Git reviews and GitHub Actions CI pipelines.

Associate Software Engineer Jul 2022 – Dec 2023
Prodapt *Hyderabad, India*

- Built full-stack web applications in Java Spring Boot and MongoDB in an Agile team using Git and Docker.

Failure Analysis Intern May 2019 – Nov 2019
Electronics Corporation of India Limited (ECIL) *Hyderabad, India*

- Debugged production Smart Energy Meter boards using test jigs, multimeter and JTAG to verify PCB hardware and firmware functionality, performed schematic-driven root-cause analysis, and fed corrective actions back to the soldering team.